Load imbalance in FESOM2/Kokkos: its origin, and the cost of removing it

6 August 2026 — Levante, 128-core CPU nodes. Per-rank per-phase wall clocks; MPI wait measured inside MPI by PMPI interposition.

The problem. At production rank counts the model spends about half of each step waiting inside MPI (51 % on CORE2 at 864 ranks, 53 % on fArc at 2048). The largest block of real computation is the ocean phase, and its per-rank cost varies by a factor of 4.8. A bulk-synchronous phase costs its slowest rank rather than its mean, so this spread is a direct tax on the step: 5.2 ms of a 48.2 ms step on CORE2 (11 %) and 15.9 ms of 84.7 ms on fArc (19 %), comparable to the entire barotropic solve.

What the imbalance is. Correlating per-rank ocean compute time against per-rank mesh content over all 2048 fArc ranks identifies it unambiguously:

correlation with per-rank ocean compute time	r
3D nodes owned ($\sum$ levels over owned columns)	**0.967**
2D nodes owned	0.003
solver wait vs. 3D nodes owned	−0.771

The imbalance is vertical. The partition balances *surface* nodes almost exactly (310–313 per rank, a spread of 1.01), but the work is per water column, and 3D nodes span **1550 to 14571, a factor of 9.4**. Ocean compute follows as 2.24 ms per 1000 3D nodes, which reproduces the measured 8.8–42.5 ms range. The negative correlation for solver wait completes the picture: deep-column ranks wait least because they are the stragglers, and the shallow ranks idle at the solve's first reduction. The imbalance is bathymetry, not sea ice.

Vertical balance depends on how the partition was weighted, and it varies across the meshes in use: NG5 and DARS are dual-weighted (3D spread 1.04–1.05), fArc and CORE2 are not (9.40, 9.60).

The test. CORE2 partitions were regenerated under 2D-only and dual 2D+3D vertex weighting from a single partitioner binary, and run in one allocation with one model binary, so the decomposition is the only variable. All arms are byte-identical on the mesh definition; only the partition files differ.

partition	3D balance	halo (nodes/rank)	s/step	vs. shipped
shipped	9.60	42	0.0440	—
2D-only weighting	9.32	42	0.0443	+0.68 %
dual 2D+3D weighting	**1.05**	**59**	0.0482	+**9.55** %

The 2D-only arm reproduces the shipped partition's imbalance and runtime, so the partitioner contributes nothing and the 9.55 % is the weighting alone.

The result. Dual weighting removes the imbalance and still costs more than it saves:

ms/step, 864 ranks	2D-only	dual	Δ
ocean compute, min / mean / max	3.8 / 13.0 / 19.4	10.1 / 14.7 / 17.5	spread 5.1 → 1.73
ocean imbalance tax (max − mean)	6.4	2.8	−3.6
ocean wait	6.7	4.9	−1.8
total compute	22.9	25.6	+2.7
solver wait	9.0	10.2	+1.2
ice-dynamics wait	4.0	5.1	+1.1
total	45.2	48.1	+**2.9**

The balancing behaves as intended — previously idle ranks take on real work and ocean wait drops 1.8 ms — but balancing the vertical requires cutting more edges, and the halo grows 40 % (42 to 59 nodes per rank). That adds 2.7 ms of compute and raises wait in the two communication-heavy phases, together exceeding the 1.8 ms recovered.

The outcome depends on decomposition, and the sign changes. The halo penalty is a constant +40 % at every rank count, but its weight in the step is not: halo rises from 9 % of owned nodes at 128 ranks to 29 % at 864, while the imbalance being removed stays roughly constant. Repeating the A/B across the ladder (min-of-2, one allocation per rung; rep-to-rep spread ≤ 0.3 %):

ranks	vertices/core	2D-only	dual	dual vs. 2D-only
256	495	0.1060	0.1011	−**4.62** %
512	248	0.0593	0.0593	0.00 %
864	146	0.0448	0.0487	+8.71 %

Conclusion. The imbalance is real, bathymetric, and worth 6.4 ms of a 45 ms step at 864 ranks. Whether removing it pays depends on how finely the mesh is decomposed: dual weighting is worth −4.6 % at 495 vertices per core, breaks even near 250, and costs +8.7 % at 146. The crossover sits close to the production point, and the gain falls inside the 300–500 vertices per core at which the model is normally run, so vertex weighting is worth applying there and worth avoiding beyond it. Both endpoints of the edgecut-against-balance trade are now measured at each rung; the dual weighting already softens its 3D criterion, so an intermediate weight remains unexplored.

One caveat. A regenerated 2D-only partition at 128 ranks made the barotropic solver diverge at the first iteration (residual 5.5×10^{45}) where the shipped and dual-weighted partitions at the same rank count both ran normally. Partition-marginal instability of this kind is a known hazard, and any new decomposition needs a stability check before use.